Romeo Velarde-Alvarez

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EDUCATION

Harvey Mudd College, Claremont, CA — B.S. Engineering, 3.87 GPA *Expected May 2028*

Relevant Coursework: Radio Frequency Circuit Design, System-on-Chip Design(Spring 2027), Analog Circuit Design(Spring 2027), Digital Electronics & Computer Engineering, Electronic & Magnetic Circuits/Devices, Experimental Engineering (Instrumentation & Sensor Data Measurement), Intro to Engineering Systems/Controls

Grading & Instruction: RF Circuit Design Lab Grader | E79 Practicum Lab Proctor | Intro to Engr Design Grader | General Chemistry Tutor & Mechanics/Wave Motion Tutor

RESEARCH EXPERIENCE

Undergraduate Researcher — Harvey Mudd College, Dept. of Engineering *Aug 2026 – Present*

Analog Circuit Engineering Lab | PI: Dr. Matthew Spencer

- Designing low-cost amplifier circuit for electropenetrography, contributing to PCB design, hardware bring-up, and testing for a compact amplifier system intended to replace bulky, noise-prone legacy EPG equipment, collaborating with 8 students to develop hw & firmware
- Developing microcontroller interfacing with EPG board's receive and transmit chains, reading the analog sensing signal via ADC and generating the stimulus voltage output on a mixed-signal PCB integrating analog front-end and digital microcontroller

Undergraduate Researcher — Harvey Mudd College, Dept. of Engineering *Jan 2026 – Present*

Cell Biology, Microfluidics, and Tissue Engineering Lab | PI: Dr. Steven Santana

- Developing electromechanical systems for printing photoinitiated hydrogel-based tissue models, extending bioprinter capability toward complex, multi-material biofabrication
- Configuring hardware–software integration for a Raspberry Pi–controlled multichannel printhead, coordinating sensor and actuator control for precision material deposition, configuring UART-USB communication between PI and printer

ENGINEERING PROJECT EXPERIENCE

Skyworks Solutions, Inc. Clinic Team Project – Gyrfalcon Wally Vector Extension *Aug 2026 – Present*

- Extending the open-source Wally RISC-V core with the RISC-V Vector (RVV) extension to add SIMD instructions for DSP, Project sponsored by Skyworks Solutions
- Contributing to micro-architecture spec and RTL design of execution unit in collaboration with team of 6 engineering students

Closed-Loop Water Level Control System – Independent Project *May 2026 – Sept 2026*

- Built water level control system by designing a microcontroller-based electronic circuit with pump & sensing with PWM actuation
- Implementing PI control to regulate flow rate of water pump under varying conditions, remote Bluetooth control with phone
- Designing PCB with KiCad, selected components and organized bill of materials

Autonomous Vertical Profiling Surface Vehicle — Experimental Engineering Team Project *Jan 2026 – May 2026*

- Collaborated in team of 4 engineering students to integrate buoyant mechanical design and electrical system in water surface robot
- Led design of sensor interface circuits w/ op-amps and potentiometers to map temperature & turbidity sensor signals to 3.3V ADC read range.

Height Gauge for Microscopy Samples — Intro to Engineering Design Team Project *Aug 2025 – Dec 2025*

- Designed a laser–photosensor detector circuit enabling precise, non-contact height measurement of electron microscope samples for external university client, collaborated in a team of 5 students to ideate, design, prototype, and test product

SKILLS

Hardware & Instrumentation: Oscilloscope, Multimeter, Function Generator, Soldering, PCB Design & Test, VNA, Data Acquisition

Software: C/C++, Python, SystemVerilog, LTspice, MATLAB, LabVIEW, Arduino IDE, SolidWorks, Quartus, KiCad

Electronics: Microcontrollers (Arduino, ESP32), Raspberry Pi, Sensor Integration, Analog & Digital I/O, PWM, Breadboard Prototyping, Signal Mapping | Feedback Control, PID Control Loops, Motor Speed Control

LEADERSHIP & WORK

Co-Founder & Co-President - Society of Hispanic Professional Engineers (SHPE) Harvey Mudd College Chapter *Dec 2025 – Present*

- Established the Mudd SHPE chapter; organize professional development workshops, industry speaker events, and networking sessions
- Secured $10,000+ in sponsorship funding to send 12 students to the SHPE National Convention

Dorm Proctor — Harvey Mudd College, Office of Residential Life *Aug 2026 – Present*

- Live-in community advisor for 40+ students; provide guidance, conflict resolution, and wellness programming; communicate resident needs to housing staff and deans

Tour Guide — Harvey Mudd College, Office of Admissions *Jan 2025 – Present*

Co-President — Society for Advancement of Latino Students at Mudd *Aug 2025 – May 2026*